STS-Based Research Directions

PBL:Hugging Face NLP Project
Team3
track 3 & session 2
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Week1 Progress

We discussed potential topics for our next steps and identified key ideas:

- **C-STS: Conditional Semantic Textual Similarity**
- **Fine-Grained Evaluation via Conditional Semantics**

Project Topic idea

Conditional Semantic Textual Similarity (C-STS)

Understanding Fine-Grained Semantic Similarities

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Reference;
C-STS: Conditional Semantic Textual Similarity - ACL Anthology
acl-org/acl-anthology: Data and software for building the ACL Anthology.

Intro: What is C-STS?

Conventional STS compresses different aspects into a single score.

C-STS, however, focuses on distinguishing key factors; For example,

Goal

(purpose/objective)

Action

(how to achieve it)

Consider the example: "open the door" vs "push the door"

- Goal: Similar (both aim to open the door) → high STS score
- Action: Different (open vs push) → different methods

C-STS performs semantic textual similarity with conditions attached (like action, goal) to capture these fine-grained differences.

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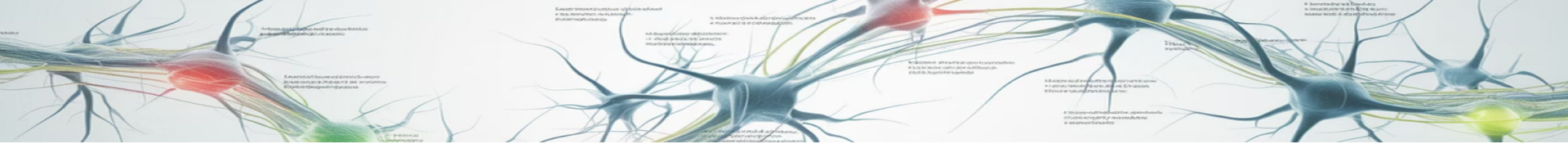

Motivation & Objective

Motivation:

High-performing models (SimCSE, Flan-T5, GPT-4) achieve less than 50% Spearman correlation on C-STS, showing that understanding fine-grained semantic similarities under conditions remains difficult.

Objective:

Propose a framework to evaluate semantic similarity conditionally by separating:

- Goal similarity (consistency of purpose)
- Action similarity (matching of actions)

This clarifies meaning structures that conventional STS cannot capture and can be applied to **text-based robot Behavior Clustering and Generation.**

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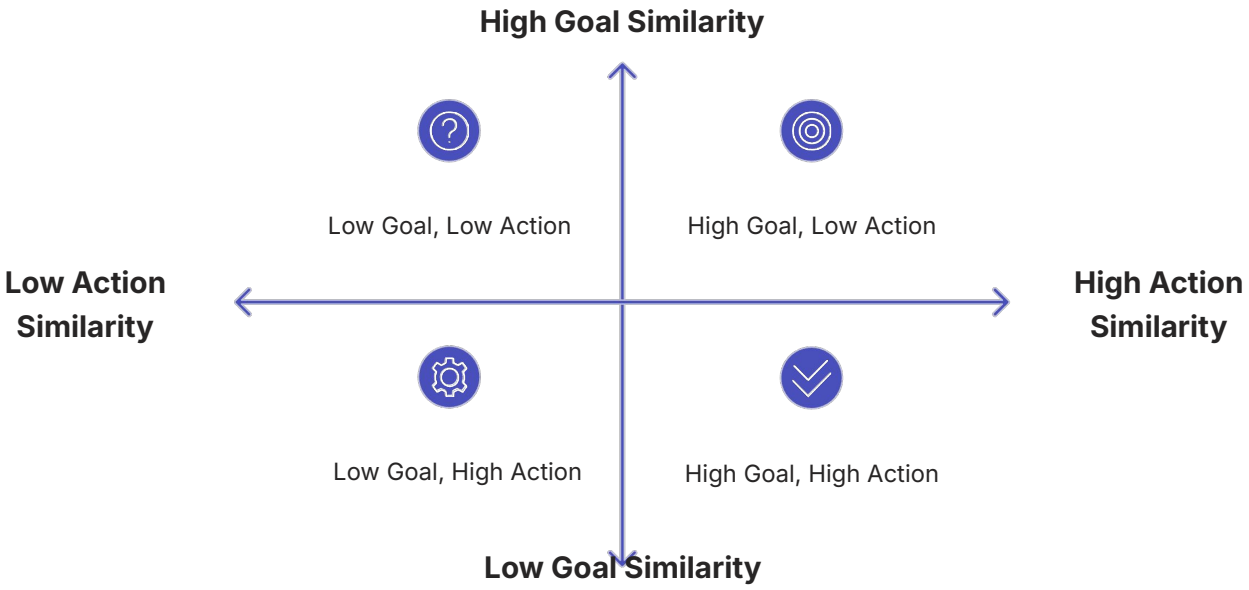
Expected Results

The proposed method will classify and visualize similarity cases:

- High goal similarity, low action similarity
- High goal similarity, high action similarity
- Other combinations



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Next Steps: Methodology

01

Data Acquisition

- Obtain STS data from Hugging Face

02

Generate Labels with Rules

- Verb extraction for action analysis
- Action similarity: based on verb agreement and similarity

03

Calculate Similarity with Models

- Baseline: Sentence-BERT (conventional STS)
- Proposed: Conditional STS model

04

Evaluation

- Comparison with conventional STS scores
- Degree of separation between goal and action
- Case study and example analysis

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Watermark Perturbation Detection

Fine-Grained Evaluation via Conditional Semantics

Presented at AI Security Conference 2024

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Problem: Limitation of Traditional Evaluation

Traditional STS

- Relies on a**single similarity score**(e.g., STS-B Spearman).
- Fails to detect subtle attacks that preserve surface semantics but break core intent (e.g., "delete" → "hide").

Our Framework

- Computes**independent scores**for key semantic dimensions (e.g., goal, action).
- Pinpoints vulnerabilities by quantifying perturbation impact on specific layers.

Key Insight: While traditional metrics focus on overall similarity, our approach provides granular visibility into**intent preservation**, enabling precise vulnerability assessment.

Core Architecture

This dual-head structure enables fine-grained evaluation by decoupling the assessment of goals from the assessment of actions.

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Execution Overview

Data Preparation

Use existing datasets (STS-B, PAWS) for training and validation.

Model Development

Fine-tune the dual-head model on the prepared data.

Evaluation

Test the model's ability to detect perturbations and quantify its performance.

Demo

Build a simple Gradio interface to showcase the framework.

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Summary

Fine-grained & Quantifiable Evaluation

This framework provides a detailed, quantifiable evaluation of watermark robustness by decomposing semantics into key dimensions (goal/action), moving beyond subjective assessment.

Addressing Critical Flaws

It effectively addresses the critical flaw of traditional single-score methods, which often fail to capture the nuanced nature of watermark attacks and defenses.

Prototype Delivery

We are committed to delivering a fully verifiable and functional prototype within the next 6 weeks to demonstrate the framework's capabilities.

Questions

idea 1: Conditional STS

- More academic, exploratory, emphasizes fine-grained semantic
- understandingImplementation might be easier (can use existing STS datasets + labeling rules)
- Will the experimental results clearly show the advantages?

idea 2: STS + Watermark Robustness

- Clear application direction, but more engineering-oriented
- Requires building a perturbation dataset and dual-task model
- Is it feasible to complete experiments in a short time?

- Which idea is more innovative?
- Which one is more suitable for this course/project?

Thank
You